

Mauricio Ernesto Rodríguez Alas

Electronic Engineer
Software Engineer

Personal Information

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Academic Background

- Phd in Computer Science - (2025-current). University of Luxembourg.
- European Master in Radiation and its Effects on MicroElectronics and Photonics Technologies (RADMEP), Erasmus Mundus (2022-2024).
- Software Engineering (Bach., 2021), Universidad Latina, Costa Rica.
- Electronic Engineering, Automation & Control (Lic., 2017), Universidad Latina, Costa Rica.
- Electronic Engineer (Bach., 2015), Universidad Latina, Costa Rica.

Work Experience

January 2021 - June 2025 - Orbital Space Technologies

- Developed an embedded system for a sub-orbital space mission.
- Architecture and develop GUI and networking for ground station segment.
- Analyzed and developed electronics, software and data collection systems for experiments.

July 2021 - March 2022 - CERN - Intern at LHCb

- Participated in lectures about detectors, particle physics and software.
- Helped improve key performance parameters on LHCb experiment software by 10-15% using Python and C++.

January 2019 - Current - Alas Mgt - Full Stack Developer and Software Lead

- Web development with Java, Angular, MySQL, MongoDB.
- Handle software projects using Agile methodologies. Develop software architectures and lead a team of 2 developers on software projects.
- Gather and perform application/business requirements with clients.

September 2020 - December 2020 - Museo Nacional de Costa Rica

- *Professional Practice* - Developed Flutter Android/iOS application to identify animals of Costa Rica in the Museum's archive using Neural Networks and image classification.

- Developed, trained, and tested an Image Classification model using machine learning with PyTorch, Keras and TensorFlow lite.

September 2018 - May 2019 - DXC Technologies: Java and IoT Intern

- IoT Prototype development. Web, Python, Arduino.
- Prototypes: Robotic Arm ordering system, IoT measurement station.

Jan 2016 - March, 2017 - Suttle Solutions CR: Laboratory and Test Engineer.

- Performed Optic Fiber Laboratory measurements and standardized testing. Elaborated lab reports and presented results.
- Test automation. Test equipment integration (LabVIEW 2017)
- Performed data analysis after experiment completion.
- Grad. Project: Climate Chamber Control System using LabVIEW 2017.

Languages

- Spanish: Native. English: C2, TOEFL iBT 2017. French: A2.

Courses/Certifications/Conferences

- ESIPAP 2022 - European School of Instrumentation Particle and Astroparticle Physics (Jan, 2022)
- Improved Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization - DeepLearningAI by Coursera (Jun, 2021).
- Deep Learning - Coursera by OpenAI. (Dec, 2020).
- Machine Learning - Coursera by Stanford University. (Oct, 2020).
- Python Bootcamp - Northwestern University. Python, Machine Learning, Classification, Time Series Analysis using Python. (Jun, 2020).
- Certified Associate in Project Management - PMI (Aug, 2020)
- LatinFusion - Workshop for Fusion energy and plasma applications, Costa Rica Institute of Technology. (Nov, 2019).
- Scrum Fundamentals Certified (Scrum Study, 2018).
- LabVIEW CLAD (2016).
- Aerospace Camp, UCR (2015).

Projects and Extracurricular Activities

- NRAO NINE Program Hub Lead (2020 - 2022) - National and International Non-traditional Exchange program participant, by National Radio Astronomy Observatory (remote due to Covid-19). Participated in theoretical and technical lectures about radio astronomy, physics and Python. Participated in the SciPy 2020 conference. Presented technical projects related to image classification.

Established radio astronomy Hub for Costa Rica as part of the NINE program. Prepared radio astronomy online course. Prepared Python, Data Manipulation, and Machine Learning online course.

- AREX - Polaris Project (Jan 2020-Dec 2021) - Science Instruments Chief. Student project to build a balloon rover on Titan. Project overview and technical R&D for the electronics team.
 - Hardwarethon (2019) (Auge UCR, ImagineXYZ) - Finalist HWThon 2019, innovation and startup MVP design, pitching ideas and prototypes in front of stakeholders and investors. Native bees conservation and education using IoT and Machine Learning.
 - Working Group 5 - ACAE (2019): Project Manager for Micro and Nano Satellites in the Region Working Group. CANSAT (Canned Satellite) Design for STEM Education: structural, electrical, control, communication, and mission design. Developed Cansat workshop for students with help of TEC SETEC Lab and TECSpace.
 - Supernovas - ACAE (2019): Instructor for STEM education, ages 8-12. Courses about rocketry, robotics, satellites and aerospace.
 - Suttle (2017) - Wind Tunnel. Part of the team that designed and constructed a small wind tunnel for up to 150 km/h for fiber products testing.
 - Suttle (2016) - Graduation Project. Design and implementation of a monitoring and control system for industrial testing equipment using LabVIEW.
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Papers, Reports and Presentations

- Mauricio Ernesto, RODRIGUEZ ALAS. Single Event Effects Instrumentation for System-on Module Testing. 2024. https://jyx.jyu.fi/jyx/Record/jyx_123456789_97356
- Project MUSA - Costa Rica Second Space Mission: Opening the Doors to Science Through Microgravity Experimentation. 29th Workshop on Space Technology for Socio-Economic Benefits. Paris, Sept. 2022. https://www.unoosa.org/documents/pdf/psa/activities/2022/UN-IAF/Posters/Mauricio_Rodriguez_Alas_-_MUSA_project_-_OST.pdf
- *Variable Scanning for Physics Efficiency Improvement in Upstream Tracking and Trackmatching Validation Algorithms in the Allen Framework*. CERN Summer Project. Nov, 2021. <https://cds.cern.ch/record/2790691>
- *MUSA: an ICE CUBES experiment using a dual culture to assess the effectiveness of microgravity effects on the TrichodermaHarzianum fungi to treat the Panama Disease*. The International Astronautical Congress (IAC). September, 2021.

- *Very Large Array Sky Survey Quicklook Image Classifier using PyTorch*. The Astronomical Data Analysis Software and Systems (ADASS). November, 2020.
<https://www.aspbooks.org/publications/532/223.pdf>
- *NINE Virtual Radio Astronomy Hub for Costa Rica and the Central America Region*. The American Astronomical Society (AAS). December, 2020.
<https://ui.adsabs.harvard.edu/abs/2021AAS...23732302R/abstract>